Prediction of Chronic Disease Risk Using Nutrition, Lifestyle & Clinical Data from The Irish Longitudinal Study on Ageing (TILDA)

Abderahman Haouit

CSIS Research Centre

Department of Computer Science and Information Systems

Faculty of Science and Engineering

University of Limerick

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1. Supervisor: Dr. Alison O'Connor
Computer Science & Information Systems
University *of* Limerick
Ireland

Abstract

Falls, frailty, and chronic conditions such as cardiovascular disease, diabetes, and obesity remain leading causes of morbidity and mortality in older adults. Yet, existing predictive models often rely narrowly on physical health indicators, overlooking nutrition and lifestyle factors that may enhance early detection of risk. This thesis develops and evaluates machine learning models using data from The Irish Longitudinal Study on Ageing (TILDA) to predict key health outcomes in a cohort of older adults. The models integrate physical, mental, lifestyle, and socio-demographic predictors, with particular emphasis on under-explored nutritional and behavioural variables highlighted in prior research. Multiple approaches are compared, including logistic regression, random forests, and gradient boosting, with performance assessed via cross-validation and feature importance analysis. By identifying high-impact, modifiable predictors, this work aims to advance early risk stratification, guide targeted prevention strategies, and inform public health policy to support healthy ageing.

Declaration

I Abderahman Haouit declare that I have produced this thesis without the prohibited assistance of third parties and without making use of aids other than those specified; notions taken over directly or indirectly from other sources have been identified as such. This thesis has not previously been presented in identical or similar form to any other Irish or foreign examination board.

The thesis work was conducted from 2025 to 2026 under the supervision of Dr. Alison O'Connor at University of Limerick.

Limerick, 2026

Acknowledgements

I would like to acknowledge my supervisor, Dr. Alison O'Connor, for her guidance, support, and for providing a well-organised environment throughout the implementation of this project.

I would also like to thank the UL Science and Engineering Research Ethics Committee for approving the ethical aspects of this research. Special thanks to the Irish Social Science Data Archive (ISSDA) team for facilitating access to the TILDA dataset, collected by Trinity College Dublin.

Your dedication goes here. Open the dedication.tex file found at:
0_frontmatter/dedication.tex

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List of Abbreviations

AI Artificial Intelligence
AUC Area Under the ROC Curve

CES-D Center for Epidemiologic Studies Depression Scale

DNN Deep Neural Network

ELSA The English Longitudinal Study of Aging
EU European Union

GBM Gradient Boosting Machine
GDPR General Data Protection Regulation

HADS Hospital Anxiety and Depression Scale
HRS The Health and Retirement Study, (a longitudinal study of ageing in the United States)

IDS The Intellectual Disability Supplement
ISSDA The Irish Social Science Data Archive

LDA Linear Discriminant Analysis

ML Machine Learning

RF Random Forest

SHAP Shapley Additive Explanations
SHARE The Survey of Health, Aging and Retirement in Europe

TILDA The Irish Longitudinal Study on Ageing

List of Abbreviations

WHO World Health Organization

Chapter 1

Introduction

Chronic conditions including cardiovascular disease, diabetes, and frailty are leading contributors to morbidity and mortality in ageing populations [1, 2]. These conditions are associated with declines in functional capacity (e.g., gait speed, grip strength [3]), increased healthcare utilisation [4], and reduced quality of life [5].

The financial and societal burden is substantial. In Europe, poor diet is a major risk factor for non communicable diseases, contributing billions in healthcare costs annually [6]. In Ireland, malnutrition alone is estimated to cost over €1.4 billion per year [7]. Recognising this, international and national policies including the World Health Organization (WHO) Global Action Plan on Noncommunicable Diseases and Ireland's *Healthy Ireland* framework highlights nutrition and lifestyle interventions as central to healthcare strategy.

In parallel, (Artificial Intelligence (AI)) and (Machine Learning (ML)) are increasingly applied in healthcare to support risk prediction and personalised intervention. However, European regulations such as the General Data Protection Regulation (GDPR) and the forthcoming European Union (EU) AI Act impose strict requirements for transparency, explainability, and governance. This regulatory context underscores the importance of interpretable and ethically robust ML approaches.

Traditional risk prediction models have largely relied on clinical biomarkers such as blood pressure and lipid profiles. Yet, growing evidence shows that broader predictors including nutrition [8], physical activity [9], mental health [10], and social determinants [11] improve predictive performance. Estimating the likelihood of developing a condition given certain risk factors, help quantify the influence of these predictors in clinical decision making.

(The Irish Longitudinal Study on Ageing (TILDA))[1] provides a nationally representative cohort of adults aged 50+ in Ireland, surveyed across multiple domains in repeated waves (waves 1-5). The raw wave datasets capture the full richness of TILDA specific measures relevant to Irish health, policy, and society. In parallel, the Harmonized TILDA dataset restructures selected variables to align with the RAND (The Health and Retirement Study, (a longitudinal study of ageing in the United States) (HRS)) framework, enabling cross national comparison with equivalent studies such as HRS (US), The English Longitudinal Study of Aging (ELSA) (UK), The Survey of Health, Aging and Retirement in Europe (SHARE) (continental Europe).

This harmonisation enhances international comparability, though at the cost of excluding some study specific variables. An additional dataset, the The Intellectual Disability Supplement (IDS), extends coverage to individuals with disabilities, improving inclusivity and reducing sampling bias. Together, these resources offer both detailed national data and a platform for internationally comparable, generalisable analyses.

Despite these advantages, existing ML research on chronic disease prediction faces persistent limitations: over reliance on clinical predictors, limited incorporation of lifestyle and social variables, predominantly cross sectional study designs that fail to capture trajectories, and a lack of interpretability consistent with emerging AI governance standards.

This thesis addresses these gaps by integrating TILDA's longitudinal, multidomain data with interpretable ML methods, demonstrating how nutrition, physical activity, and mental health can be combined with traditional clinical markers to improve prediction of ageing related outcomes.

[1]TILDA data are pseudonymised and accessed under licence via The Irish Social Science Data Archive (ISSDA). Use adheres to UL S&E ethics approval and data governance requirements.

1.1 Research Problem and Motivation

Despite increasing policy interest in lifestyle and nutrition and advances in ML methods, approaches to chronic disease prediction in older adults remain limited in several ways.

- **Limited Integration of Multidimensional Data:** Most models prioritise clinical variables while underutilising modifiable lifestyle and nutrition factors [8, 12].

- **Underuse of Longitudinal Dynamics:**Few studies leverage repeated measures to capture developments of risk, such as frailty progression [13]. Probabilistic reasoning can help estimate the likelihood of developing a condition at a future time given earlier measurements, for example calculating $P(\text{Frailty at wave 5} \mid \text{Gait speed at wave 1}, \text{Nutrition at wave 2})$.

  This approach helps quantify how early indicators influence later outcomes and supports better feature selection and timing of interventions in predictive models.

- **Lack of Explainability:**Many ML models are not interpretable, limiting clinical adoption and failing to meet emerging regulatory requirements for trustworthy AI in healthcare [14].

Addressing these challenges is essential to improve predictive accuracy, support evidence based interventions, and ensure alignment with evolving policy and regulatory priorities.

1.2 Research Aims and Objectives

The goal of this thesis is to develop and validate ML models that integrate nutrition, lifestyle, and mental health features with clinical data to improve prediction of chronic disease and ageing related outcomes in a representative cohort of older adults. This goal will be achieved through the following objectives:

- Review existing applications of ML for chronic disease prediction in ageing populations, with emphasis on gaps in lifestyle and nutrition integration.
- Preprocess and harmonise TILDA waves 1-5, and engineer multidomain features (nutrition, activity, mobility, mental health, clinical indicators).
- Develop and compare baseline statistical models (e.g., logistic regression) with advanced ML methods (Random Forest (RF), Gradient Boosting Machine (GBM)).
- Evaluate models using discrimination (Area Under the ROC Curve (AUC)), calibration, and error metrics (accuracy, precision, recall, F1-score), perform ablation analyses to assess the incremental value of lifestyle and nutrition predictors.
- Identify actionable predictors (e.g., vitamin D status, gait speed reserve) and highlight modifiable risk factors for intervention.
- Document ethical, governance, and reproducibility practices for handling longitudinal pseudonymised health data.

1.3 Contributions of the Research

This research makes contributions at three levels:

- **Empirical:** Demonstrates how nutrition, lifestyle, and mental health features enhance prediction when combined with established clinical baselines.
- **Methodological:** Provides a reproducible pipeline for harmonising longitudinal TILDA data and integrating multidomain features into ML workflows.
- **Clinical:** Identifies modifiable predictors with relevance for targeted interventions and policy to promote healthy ageing.

1.4 Thesis Outline

In *Chapter 2* a literature review on ML prediction of chronic disease in ageing populations, with emphasis on lifestyle and nutrition predictors and methodological considerations is presented. *Chapter 3* discusses the TILDA study design, data preprocessing, harmonisation, and feature engineering (Flow Charts...). In *Chapter 4* the modelling framework is outlined. This includes baseline and advanced ML, hyperparameter tuning, validation, and evaluation metrics. *Chapter 5* shows the model result, including model comparisons, assessments of feature importance, and an ablation analyses. In *Chapter 6* the implications, limitations, ethical considerations are discussed, and future directions for the research proposed. Additional supporting documents are included in appendices.

Chapter 2

Literature Review

This chapter reviews literature on ageing, chronic disease, and predictive modelling using TILDA. The focus is on three domains central to later life health: frailty and physical function, mental health, and nutrition and biological ageing. Chronic conditions including multimorbidity, frailty, and mental health difficulties are leading causes of morbidity and reduced quality of life in older adults, often overlapping to accelerate functional decline. In line with the WHO healthy ageing framework, research using TILDA highlights frailty, mental health, and nutrition as interconnected determinants of later life outcomes. The following subsections examine each domain in turn.

2.1 Frailty and Physical Function

Frailty is a reflection of higher vulnerability and declining physiological reserve. In TILDA, it is captured through both the frailty phenotype and the frailty index, with gait speed, grip strength, and mobility decline serving as practical predictors. These measures strongly predict hospitalisation and mortality.

2.1.1 Probabilistic Interpretation of Frailty

One can argue that from a probabilistic perspective, the chance of an adverse outcome (such as mortality, Y) given observed features (such as frailty markers, X) as a conditional probability $P(Y \mid X)$. This allows us to deduce which frailty features are most predictive of adverse outcomes. Features may be independent, dependent, or mutually exclusive.

For instance, presence or absence of metabolic syndrome represents a mutually exclusive event that can inform risk models.

$$P(X_i \cap X_j) = P(X_i)P(X_j)$$

for independent features X_i and X_j, guiding feature selection and reducing redundancy in model inputs. This reflects the likelihood that an individual with certain frailty characteristics will experience a particular outcome.

Evidence suggests that frailty is dynamic, not irreversible. For example, multi state modelling showed that many prefrail adults transition back to non frail, although established frailty strongly predicts mortality [15]. Metabolic syndrome, on the other hand, speeds up the onset of frailty, which suggests a metabolic pathway of vulnerability. [16]. The presence of metabolic syndrome modifies the conditional probability $P(\text{Frailty} \mid \text{Metabolic Syndrome})$ and may increase the likelihood of adverse outcomes over time. Another detail is that age matters: obesity predicts mobility decline earlier in life, whereas later in life, traits that protect against this, including grip strength, become more significant [17].

These findings partly conflict: frailty appears both reversible and progressive, depending on risk context. This complexity highlights the need for probabilistic models that can update beliefs about risk as new features are observed, as formalised by Bayes' theorem:

$$P(Y \mid X) = \frac{P(X \mid Y)P(Y)}{P(X)}$$

where $P(Y \mid X)$ is the posterior probability of outcome Y given features X. This enables us to incorporate prior knowledge and observed data, improving the interpretability of feature outcome relationships in ageing. Here, $P(Y \mid X)$ is the posterior probability = (Likelihood * Prior / Evidence), representing the updated belief about the outcome after observing features. $P(Y)$ is the prior probability, reflecting baseline risk. $P(X \mid Y)$ is the likelihood, quantifying the probability of observed features given the outcome, and $P(X)$ is the evidence (the overall probability of the features). This framework allows us to update predictions as new data is observed, improving interpretability in feature outcome relationships.

Limitations remain. Causal inference is limited by observational designs, and predictors are frequently examined separately, ignoring interactions. Most frailty models have only been with the TILDA dataset and with this present study focuses on the same TILDA but with an expansion on the harmonised set to explore more longitudinal prediction within this context. These gaps point to the need for multidimensional approaches. Traditional regression models may not fully capture age dependent interactions, suggesting a role for more flexible methods such as machine learning techniques and models.

2.2 Mental Health in Older Adults

Mental health, including depression, anxiety, loneliness, and social isolation, strongly shapes quality of life. Evidence from TILDA shows strong links to both physical and social factors. [18] found that participation in physical, social, and cognitive activities summarised by the Act-Belong-Commit (ABC) indicator was associated with lower depression and anxiety. This suggests social integration and lifestyle protect mental wellbeing. However, [19] reported that self reported measures may exaggerate such associations..., raising concerns of bias. These two findings conflict: the effect of social engagement is strong when self reported, weaker when objectively assessed.

Objective data strengthen the case for activity. Accelerometer based measures show a dose response link between moderate to vigorous activity and lower depressive symptoms [9]. Yet reverse pathways are also evident: incident falls and functional decline predict later depression, refereed by fear of falling [20]. Together, these results reveal a bidirectional relationship, mental health both influences and is influenced by physical decline.

Most studies rely on broad self report scales such as Center for Epidemiologic Studies Depression Scale (CES-D) or Hospital Anxiety and Depression Scale (HADS), with little biomarker or clinical validation. Samples remain heavily Irish based and 50+ in age, limiting wider relevance.

2.2.1 Probabilistic Modelling of Mental Health Outcomes

This literature suggests mental health in ageing cannot be modelled as a one way effect. It requires longitudinal, multidimensional approaches that can capture mutual pathways, an area where machine learning is well suited especially when combined with probabilistic models that estimate the likelihood and posterior probability of mental health outcomes given multiple interacting features.

For example, we can estimate:

$$P(\text{Depression} \mid \text{Activity, Falls, Social Engagement}),$$

quantifying how these interacting features together influence mental health outcomes. Probabilistic modelling helps determine which features have the strongest effect and which may be redundant for predictive purposes.

2.3 Nutrition and Biological Ageing

Nutrition is increasingly recognised as a modifiable factor in ageing and frailty. Biomarkers like vitamin D, lutein, and zeaxanthin have been linked to physical and cognitive outcomes in TILDA. [8] showed that higher plasma lutein and zeaxanthin dietary carotenoids found in leafy vegetables were linked to slower frailty progression. By contrast, [21] found vitamin D deficiency predicted diabetes and musculoskeletal decline, especially among adults with low sunlight exposure. These studies point to different nutritional mechanisms: antioxidant protection versus metabolic regulation.

2.3.1 Bayesian Perspective on Nutritional Biomarkers

The effect of nutrition on ageing outcomes can be formalised probabilistically using Bayes theorem:

$$P(\text{Outcome} \mid \text{Nutritional Status}) = \frac{P(\text{Nutritional Status} \mid \text{Outcome})P(\text{Outcome})}{P(\text{Nutritional Status})},$$

where the likelihood function $P(\text{Nutritional Status} \mid \text{Outcome})$ quantifies how observed nutritional biomarkers relate to health states, and the posterior $P(\text{Outcome} \mid$

Nutritional Status) captures updated beliefs about outcomes, including uncertainty in small-sample studies. This helps identify which biomarkers are most informative for predicting ageing outcomes which supports feature selection in ML models.

2.3.2 Contextual and Integrative Considerations

Malnutrition also reflects social and functional risks. [2] found that hospitalisation, functional loss, and poor social support increased malnutrition risk, with notable sex differences. This highlights a limitation in much nutritional research: biological markers are studied without integrating social context. The difference between biological and chronological age is a major point of debate. Biomarker based metrics that better capture these cumulative exposures, like frailty indices or epigenetic clocks, frequently vary from chronological age. McCarthy and Azizi link metabolic syndrome and nutritional deficits to accelerated biological ageing [22, 23]. Although the majority of the evidence is cross sectional and has little ability to capture changes over time, these indicators offer a more precise prediction of vulnerability. Reliability is also decreased in biomarker research with small sample sizes, which can be formalised through the likelihood function $P(\text{Data} \mid \theta)$ and its impact on posterior uncertainty in Bayesian inference.

Taken together, nutrition interacts with biological, metabolic, and social systems. Isolating single nutrients risks oversimplification. Integrative modelling that combines biomarkers, clinical factors, and social determinants supported by machine learning pipelines offers a more promising route. Models that estimate the joint distribution of features and outcomes can better capture the complex dependencies relevant to biological ageing.

2.4 Machine Learning in Ageing and Chronic Disease

ML is increasingly applied in ageing research because it can capture non linear and multidimensional relationships that traditional regression models, such as logistic regression,often miss [24, 25].

2.4.1 Feature Sets in ML Studies

Studies using ML in ageing research include many types of predictors: clinical measures, sociodemographic factors, lifestyle habits, cognitive tests, and psychosocial variables. A recurring pattern, however, is that models often underline easily measured physical health data, such as comorbidity counts, mobility scores, or vital signs, while more complex but important predictors, including nutrition and social factors, are underrepresented [26]. This imbalance may limit the ability of models to capture the full complexity of ageing, especially considering that mental health and nutrition exert independent and interacting effects on later life outcomes see (here). Models that focus only on readily available clinical data risk reproducing the same biases seen in traditional regression approaches.

2.4.2 ML Models Used

The methodological approaches used in ageing research span both traditional and advanced models. Logistic regression and Markov models remain common baselines due to their interpretability and simplicity, yet they are often unable to capture non linearities or higher order interactions. More recent work has shifted towards ensemble methods such as random forests (RF) and gradient boosting machines (GBM), as well as deep neural networks (Deep Neural Network (DNN)), which generally achieve higher predictive accuracy.

Similarly, [27] evaluated mortality prediction using a standard logistic regression model with demographic, clinical, and functional predictors, achieving an AUC of 0.78, whereas, ensemble methods in [24] reached 0.854 for diabetes prediction, showing how flexible, non linear models can better capture complex interactions among ageing related predictors.

2.4.3 Performance and Validation

Predictive performance across studies is generally moderate to strong, with AUC values between 0.70 and 0.90 for outcomes like diabetes, gait speed, depression, and mortality. This indicates that models can distinguish reasonably well between individuals who will or will not experience an outcome.

However, there are important limitations. Many studies rely only on internal validation, testing models on the same data used for training. This can inflate performance estimates and reduce confidence in applying the model to new populations. To add, Most studies are cross sectional, so causal or temporal effects cannot be assessed. When combined with the underuse of nutrition and social predictors, these issues indicate that current models may not fully represent the multidimensional nature of ageing and could produce context specific results rather than generalisable predictions [26].

2.4.4 Explainability in ML for Ageing

As ML models become more complex, interpretability emerges as a critical challenge. Explainability tools like Shapley Additive Explanations (SHAP) can highlight feature importance, but their use is inconsistent, particularly in deep learning models [3, 24, 28]. These methods improve transparency and can highlight unexpected drivers of outcomes, such as psychosocial variables being stronger predictors than biomedical ones, yet black box concerns remain, especially for deep learning models, where feature interactions are difficult to track. For ageing research, where clinical adoption requires both accuracy and trust, explainability is not optional but essential. Current efforts show promise but remain fragmented, with no consensus on best practice. This gap links directly to the broader methodological limits discussed in the next section on gaps in the literature.

2.5 Gaps in the Literature

First, most studies rely on TILDA or similar cohorts of Irish and predominantly older adults (50+). This limits confidence in how well results transfer to other populations. The present project focuses on TILDA to study ageing in depth but also makes use of the harmonised dataset, which links TILDA with comparable UK and US cohorts. This helps place findings in a broader international context while retaining a core emphasis on older adults.

Second, there is an over reliance on physical health predictors such as comorbidity counts and mobility measures, while nutrition, social, religious or spiritual, and environmental variables are often overlooked. As shown in [8], nutritional biomarkers like lutein and zeaxanthin are strong predictors of frailty, yet these are rarely integrated into broader ML pipelines. By contrast, studies such as [2] highlight the role of social support and functional decline in malnutrition risk. Together, these examples show that excluding nutrition and social factors risks producing models that reinforce the same biomedical biases seen in regression heavy work. From a Bayesian viewpoint, the omission of key features reduces the explanatory power of the model and weakens the posterior probability $P(\text{Outcome} \mid \text{All Features})$ compared to $P(\text{Outcome} \mid \text{Limited Features})$. Including a richer set of features increases the likelihood that the model accurately reflects the true data generating process.

Third, methodological choices remain conservative. Regression models still dominate, while more advanced approaches such as deep neural networks or transformers are rarely explored. For instance, [27] achieved an AUC of 0.78 using logistic regression for mortality prediction in TILDA, whereas [24] reported an AUC of 0.854 with ensemble methods for diabetes incidence. This contrast demonstrates both the dominance of regression and its weaker performance compared to flexible ML models. Where ensemble or brain age approaches have been attempted, validation is almost always internal, with little cross cohort testing, inflating performance estimates and undermining clinical translation. Weak validation and limited feature sets reduce the generalisability of the estimated conditional probabilities and increase the risk of overfitting. Maximum likelihood estimation and Bayesian inference can provide a formal framework for assessing model fit and updating predictions as more diverse and representative data become available.

Fourth, many outcomes remain underexplored. Studies on dementia or frailty frequently focus on cognitive or physical decline while leaving aside outcomes critical to older adults, such as pain, sleep quality, and subjective quality of life. For example, [8] used cross sectional plasma biomarkers to study frailty but did not assess sleep or pain, while gait speed models such as [3] similarly neglect subjective quality of life. This shows how reliance on narrow outcomes leaves broader aspects of wellbeing unmeasured.

Fifth, cross sectional bias and data handling weaken reliability. Much nutritional work, such as [8], relies on single time point biomarker data, which cannot capture longitudinal change. By contrast, studies applying multi wave TILDA data [15] demonstrate the added value of modelling temporal dynamics. Similarly, missing data are often handled through list wise deletion, as in [27], which risks biasing results toward healthier participants, whereas more robust imputation approaches remain underused. Missing or poorly handled data can distort the likelihood function and hence the posterior predictions, reducing confidence in the model's outputs.

Finally, interpretability challenges persist. While tools like SHAP have been applied to highlight feature importance [24, 28], their use is inconsistent, and consensus on best practice has yet to emerge. Without transparent explanations, even accurate models may face barriers to clinical acceptance. Taken together, these examples show that ageing research continues to rely on narrow predictors, conservative models, and weak validation. Addressing these gaps requires integrative, diverse, and transparent approaches that link multidimensional predictors with explainable ML.

2.6 Chapter Summary

This chapter reviewed evidence on ageing and chronic disease across three domains, frailty and physical function, mental health, and nutrition and biological ageing. We also examined how machine learning has been applied to predict later life outcomes. The findings showed that while frailty is partly reversible, it remains strongly tied to mortality; that mental health both shapes and is shaped by physical decline; and that nutrition interacts with biological, metabolic, and social systems in ways that are often underestimated.

While machine learning has improved prediction accuracy in ageing research, challenges remain in integrating diverse predictors and ensuring interpretability and longitudinal validation.

This study builds on previous research by leveraging TILDA alongside the harmonised dataset linking UK and US cohorts, enabling longitudinal prediction of ageing outcomes. The main contribution lies in integrating nutrition, lifestyle, and clinical predictors, applying temporal modelling, and emphasising interpretability, addressing key limitations identified in prior work. These directions motivate the methodological choices outlined in Chapter 4.

Chapter 3

Data and Preprocessing

3.1 Insert a Figure

We refer to a figure in the text like this: (Figure 3.1).

Figure 3.1: Title of Figure - Description. [Source: (**?**)]

3.2 Creating a list

- **(2000)** item 1.
- **(2004)** item 2.
- **(2010)** item 3.
- **(2013)** item 4.

3.2.1 Creating a Table

Table 3.1: Table Title

	Resolution	Min	Max
Gyroscope	4.5mV / °/s	0.27 °/s	406 °/s
Accelerometer	600mV /g	0.002g	2g
Magnetometer	385mV/ gauss	0.317 gauss	6 gauss

3.3 Quote

An in-text quote is declared in the following way:

> Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! (CommonSense, p. 0)

Another way of doing it is:

There are in our existence spots of time,
That with distinct pre-eminence retain
A renovating virtue, whence-depressed
By false opinion and contentious thought,
Or aught of heavier or more deadly weight,
In trivial occupations, and the round
Of ordinary intercourse-our minds
Are nourished and invisibly repaired;
A virtue, by which pleasure is enhanced,
That penetrates, enables us to mount,
When high, more high, and lifts us up when fallen.
(**?** , verses 208-218)

Chapter 4

Methodology

4.1 Math

In-text Math

- A vector vector $\hat{A}_{ba}\{x_{ba}, y_{ba}, z_{ba}\}$
- Another vector $R^b\{\hat{t}_{1b}, \hat{t}_{2b}, \hat{t}_{3b}\}$ as:

A series of Equations:

$$\hat{t}_{1b} = \hat{A}_{ba} \tag{4.1}$$

$$\hat{t}_{2b} = \frac{\hat{A}_{ba} \times \hat{M}_{bm}}{\mid \hat{A}_{ba} \times \hat{M}_{bm} \mid} \tag{4.2}$$

$$\hat{t}_{3b} = \hat{t}_{1b} \times \hat{t}_{2b} \tag{4.3}$$

$$R^a = R^b(R^i)^T = [\hat{t}_{1b}, \hat{t}_{2b}, \hat{t}_{3b}][\hat{t}_{1i}, \hat{t}_{2i}, \hat{t}_{3i}]^T \tag{4.4}$$

where the symbol ^{T} denotes transposition.

$$\hat{t}_{2b} = \frac{\hat{A}_{ba} \times \hat{M}_{bm}}{\mid \hat{A}_{ba} \times \hat{M}_{bm} \mid} \tag{4.5}$$

$$= \frac{(y_{ba}z_{bm} - y_{bm}z_{ba}, z_{ba}x_{bm} - z_{bm}x_{ba}, x_{ba}y_{bm} - x_{bm}y_{ba})}{\mid A_{ba} \mid\mid M_{bm} \mid \sqrt{1 - (\hat{A}_{ba} \cdot \hat{M}_{bm})^2}}$$

$$= \frac{-0.2338, -0.0931, -0.0651}{0.2601}$$

$$= (-0.8989, -0.3579, -0.2503)$$

which if normalised gives:

$$= (-0.8995, -0.3581, -0.2504)$$

A matrix

$$R^b = [\hat{t}_{1b}, \hat{t}_{2b}, \hat{t}_{3b}] = \begin{pmatrix} -0.2698 & -0.8995 & 0.3439 \\ 0.0035 & -0.3581 & -0.9337 \\ 0.9629 & -0.2504 & 0.0997 \end{pmatrix}$$

Chapter 5

Results

5.1 References

Open the file 'references_myphd' with BibTex. The file is located in '/7_references/'

The file contains a series of templates for the proper formatting of references according to the Harvard referencing style. Click on each reference in the text below to see how Latex has formatted the reference on the References section.

- **Book**: (**?** , p. 10) (page number required only if referring to an in-text quote)
- **Book Chapter**: (**?**)
- **Book Contribution**: (**?**)
- **MUSIC CD/DVD**: (**?**)
- **Journal Article**: (**?**)
- **Patent**: (**?**)
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- **Unpublished Report**: (**?**)
- **Web Videos**(**?**) (. . . not the name of the person who uploaded the video though)
- **Web Paper**: (**?**)
- **Web Journal Article**: (**?**)
- **Webpage**: (**?**)
- **Downloaded Images or Sounds**: (**?**)

You can also you this kind of referencing if needed in the text:
Surname [**?**] said that . . .

Chapter 6

Conclusions and Future Directions

6.1 Summary

6.2 Conclusions

6.3 Contributions

6.4 Future Work

Appendix A

Insert a figure

Appendix B

Title

Type here you text as usual . . .

Appendix C

Code for estimating attitude

AHRS_TRIAD.C

```
/*
 *
 *  Copyright (c) 2013, Giuseppe Torre
 *  All rights reserved.
 *
 *  Redistribution and use in source and binary forms, with or without
 *  modification, are permitted provided that the following conditions are met:
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 *  ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT
 *  (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS
 *  SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.
 *
 */



#include "ext.h"
#include "ext_mess.h"
#include <string.h>
#include <stdio.h>
#include <math.h>
#define EPSILON 1e-6

//*****************CALIBRATION STUFF************************//
/* INTRODUCE THE OFFSET VALUES IN THE FOLLOWING VARIABLES IN ADC values */
#define ACCELX_OFFSET 2043 //initial offset in accelerometer X
#define ACCELY_OFFSET 2053 // initial offset in accelerometer Y
#define ACCELZ_OFFSET 2264//initial offset in accelerometer Z

#define MagX_OFFSET 1917 //initial offset in Magnetometer X
```

```
#define MagY_OFFSET 1813 // initial offset in Magnetometer Y
#define MagZ_OFFSET 1667//initial offset in Magnetometer Z
/* WE SHOULD CALIBRATE THE FOLLOWING VALUES FOR EACH PARTICULAR IMU axis */
#define ACCELX_Resolution 536
#define ACCELY_Resolution 661
#define ACCELZ_Resolution 559
#define MagX_Resolution 186
#define MagY_Resolution 189
#define MagZ_Resolution 170
//****************** END CALIBRATION STUFF  *************************//


void *this_class; // Required. Global pointing to this class

typedef struct _triad // Data structure for this object
{
        t_object m_ob;    // Must always be the first field; used by Max

        Atom m_args[9];   // we want our inlet to be receiving a list of 10 elements

        long m_value;     //inlet

        void *m_R1;                //these are all the outlets for the 3 X 3 Matrix
        void *m_R2;                // R1 --> firts top left cell ..R2 middle cell of the first
            row ...and so on
        void *m_R3;                                     //
        void *m_R4;                                     // End Outlets
            ________________________________________
} t_triad;

                        void *triad_new(long value);


                        void triad_assist(t_triad *triad, void *b, long msg, long arg, char *
                            s);
                        void triad_free(t_triad *triad);



                        void triad_list(t_triad *x, Symbol *s, short argc, t_atom *argv);




                        void MatrixByMatrix(double *Result,double *MatrixLeft,double *
                            MatrixRight);

                        void Matrix2Quat(double *Quat,double *Matrix);
                        void Quat2Matrix(double *Matrix, double *Quat);
                        void inverseQuat(double *InvQuat, double *RegQuat);
                        void NormQuat(double *YesQuat, double *NotQuat);
                        void Slerp(double *NewQuat, double *OldQuat, double *CurrentQuat);
                        void NormVect(double *YesVect, double *NotVect);
                        double orientationMatrix[9];
                        double Result[9];
                        int    i;
                        double InvorientationMatrix[9];

                        double temp[6];
                        double ref[6];
                        double vectAx[3];
                        double vectAy[3];
                        double vectAz[3];
                        double vectBx[3];
                        double vectBy[3];
                        double vectBz[3];
                        double MagnCrosProd_A;
                        double MagnCrosProd_B;
                        double accnorm, magnorm, earthnorm, VectAynorm, VectAznorm,
                            VectBynorm, VectBznorm;
```

```
                              double m[9], n[9];
                              double quat_e[4];
                              double invquat_e[4];
                              double mult;
                              double quat_new[4];
                              double quat_old[4];
                              double ecs,accex_ADCnumber,y,accey_ADCnumber,z,accez_ADCnumber,mx,
                                   magnx_ADCnumber,my,magny_ADCnumber,mz,magnz_ADCnumber;
// SLERP Variables
                              double trace, Suca;
                              double tol[4], omega, sinom, cosom, scale0, scale1, tez,
                                   orientationMatrixA[9];

int main(void)
{
          // set up our class: create a class definition
          setup((t_messlist**) &this_class, (method)triad_new, (method)triad_free, (short)
               sizeof(t_triad), 0L,A_GIMME, 0);
     addmess((method)triad_list, "list", A_GIMME, 0);
          addmess((method)triad_assist, "assist", A_CANT, 0);
          finder_addclass("Maths","triad");
          post("....I'm-TRIAD-Object!....from-AHRS-Library...",0);
          return 0;
}

/* -------------triad_new ---------------*/

void *triad_new(long value)
{
          t_triad *triad;
          triad = (t_triad *)newobject(this_class);          // create the new instance and return
                 a pointer to it
          triad->m_R4 = floatout(triad);
          triad->m_R3 = floatout(triad);
          triad->m_R2 = floatout(triad);
          triad->m_R1 = floatout(triad);

return(triad);
}

etc.

etc.
```

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